CLASSWORK 10

Be Bayesian

CPE 486/586

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Due: October 19, 2026, In-Class

20 points

Last month, a small engineering office logged the 100 emails that reached its mail server. A staff member read them and labeled the messages spam ($Y = 1$) or awesome ($Y = 0$), where awesome means legitimate mail. The filter tracks two binary features.

- $X_1 = 1$ when the email contains the word "offer", and $X_1 = 0$ otherwise.
- $X_2 = 1$ when the email contains the word "attached", and $X_2 = 0$ otherwise.

Table 1 lists the counts. Read it one row at a time. The row $(1, 0)$, for instance, says that 12 spam emails and 18 awesome emails contained "offer" without "attached".

Table 1. Counts of the 100 labeled emails.

X_1 ("offer")	X_2 ("attached")	Spam ($Y = 1$)	awesome ($Y = 0$)	Row total
1	1	24	6	30
1	0	12	18	30
0	1	0	30	30
0	0	4	6	10
Column total		40	60	100

Part 1: Priors and Likelihoods from Counts (4 points)

(a) **[1 point]** Estimate the priors from the column totals of Table 1.

$$P(Y=1) = _____ \qquad P(Y=0) = _____$$

(b) **[2 points]** Fill in Table 2 with the remaining class-conditional joint likelihoods $P(X_1 = x_1, X_2 = x_2 \mid Y = y)$. Divide a count by its class total. The spam column must add to 1, and so must the awesome column.

Table 2. Joint likelihoods.

x_1	x_2	$P(x_1, x_2 \mid Y=1)$	$P(x_1, x_2 \mid Y=0)$
1	1	$24/40 = 0.6$	
1	0		
0	1		
0	0		
Column sum			

(c) **[1 point]** Apply the sum rule to Table 2 and fill in Table 3. As a pattern, $P(X_1 = 1 \mid Y = y) = P(1,1 \mid Y = y) + P(1,0 \mid Y = y)$.

Table 3. Marginal likelihoods of the two words.

	$y = 1$ (spam)	$y = 0$ (awesome)
$P(X_1 = 1 \mid Y = y)$	$0.6 + 0.3 = 0.9$	
$P(X_2 = 1 \mid Y = y)$		

Part 2: Full Bayes with the Joint Likelihood (5 points)

Two new emails arrive. Email A contains "offer" and lacks "attached", so $\mathbf{x}_A = (1, 0)$. The second one, Email B, lacks "offer" and contains "attached", so $\mathbf{x}_B = (0, 1)$. In this part you'll take the likelihoods straight from Table 2.

(a) **[4 points]** Complete Table 4.

Table 4. Posteriors under the joint model.

Quantity	Email A, $\mathbf{x}_A = (1, 0)$	Email B, $\mathbf{x}_B = (0, 1)$
$P(\mathbf{x} \mid Y = 1)\, P(Y = 1)$	$0.3 \times 0.4 = 0.12$	
$P(\mathbf{x} \mid Y = 0)\, P(Y = 0)$		
$P(\mathbf{x})$ by the sum rule		
$P(Y = 1 \mid \mathbf{x})$		
$P(Y = 0 \mid \mathbf{x})$		
Decision ω^* (spam or awesome)		
$P(\text{error} \mid \mathbf{x})$		

(b) **[1 point]** Count the emails in Table 1 whose features equal $\mathbf{x}_A$ and divide by 100. Compare the result with $P(\mathbf{x}_A)$ in Table 4, then explain the agreement in one sentence.

Observed frequency of $\mathbf{x}_A$: _____

Part 3: Naive Bayes (5 points)

Naive Bayes replaces the joint likelihood with a product of marginals,

$$P(\mathbf{x} \mid Y = y) \approx P(X_1 = x_1 \mid Y = y)\, P(X_2 = x_2 \mid Y = y).$$

Take the factors from Table 3. When a word is absent, use $P(X_j = 0 \mid Y = y) = 1 - P(X_j = 1 \mid Y = y)$.

(a) **[4 points]** Complete Table 5 for Email A and Email B.

Table 5. Posteriors under naive Bayes.

Quantity	Email A, $\mathbf{x}_A = (1,0)$	Email B, $\mathbf{x}_B = (0,1)$
$P(x_1 \mid Y=1)\,P(x_2 \mid Y=1)$	$0.9 \times 0.4 = 0.36$	
$P(x_1 \mid Y=0)\,P(x_2 \mid Y=0)$		
Spam term, times $P(Y=1)$		
awesome term, times $P(Y=0)$		
$P_{\text{NB}}(\mathbf{x})$ by the sum rule		
$P(Y=1 \mid \mathbf{x})$		
$P(Y=0 \mid \mathbf{x})$		
Decision ω^* (spam or awesome)		
$P(\text{error} \mid \mathbf{x})$		

(b) **[1 point]** Write down the class that the joint model picks for Email A and the class that naive Bayes picks for it.

Joint model: _____ Naive Bayes: _____

Part 4: Testing the Naive Assumption (2 points)

(a) **[1 point]** Conditional independence requires

$$P(X_1=1, X_2=1 \mid Y=y) = P(X_1=1 \mid Y=y)\,P(X_2=1 \mid Y=y).$$

Check this equation for $y=1$, then for $y=0$, with numbers from Tables 2 and 3.

Class	Left side (Table 2)	Right side (Table 3)	Equal?
$y=1$ (spam)	0.6		
$y=0$ (awesome)			

(b) **[1 point]** Compare $P_{\text{NB}}(\mathbf{x}_A)$ in Table 5 with the observed frequency from Part 2(b). Explain the gap in one or two sentences.

Part 5: Interpretation (4 points)

(a) **[1 point]** The joint model reports $P(Y = 1 \mid \mathbf{x}_B) = 0$. Which count in Table 1 produced this zero? Would you deploy a filter that is certain Email B is awesome? Say which posterior for Email B you'd report, and give a reason.

(b) **[1 point]** Email C contains "offer" and "attached", so $\mathbf{x}_C = (1, 1)$. Compute $P(Y = 1 \mid \mathbf{x}_C)$ under the joint model and under naive Bayes.

Joint: _____ Naive: _____

Naive Bayes gives Email C the posterior it gave Email A. Use Table 3 to explain why.

(c) **[1 point]** Suppose the office installs the same filter in a department where spam makes up 10% of incoming mail, so $P(Y = 1) = 0.1$. Keep the likelihoods of Table 2. Recompute $P(Y = 1 \mid \mathbf{x}_C)$ under the joint model and state the new decision.

$P(Y = 1 \mid \mathbf{x}_C) =$ _____ Decision: _____

(d) **[1 point]** Bayes' rule in odds form reads

$$\frac{P(Y=1 \mid \mathbf{x})}{P(Y=0 \mid \mathbf{x})} = \frac{P(Y=1)}{P(Y=0)} \times \frac{P(\mathbf{x} \mid Y=1)}{P(\mathbf{x} \mid Y=0)}.$$

With the original prior $P(Y=1) = 0.4$, write the posterior odds for Email C under the joint model as prior odds times likelihood ratio. Then evaluate the sigmoid at $z = \ln(\text{posterior odds})$ and compare the result with Part 5(b).

Prior odds: $0.4/0.6 = 2/3$

Likelihood ratio: _____

Posterior odds: _____

$g(\ln(\text{posterior odds}))$: _____

Bonus: From Naive Bayes to Logistic Regression (+4 points, optional)

For binary features, naive Bayes produces log posterior odds that are linear in the features,

$$\ln \frac{P(Y=1 \mid \mathbf{x})}{P(Y=0 \mid \mathbf{x})} = w_0 + w_1 x_1 + w_2 x_2, \qquad w_j = \ln \frac{p_{j1}}{p_{j0}} - \ln \frac{1-p_{j1}}{1-p_{j0}},$$

where $p_{jy} = P(X_j = 1 \mid Y = y)$.

(i) Show that $w_2 = 0$ for this data set.

(ii) Compute the joint-model posterior for $\mathbf{x} = (0,0)$. Then argue that a logistic regression with $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ cannot reproduce the joint-model posteriors of the four feature patterns. Hint: compare the change in log odds when x_1 switches on with $x_2 = 0$, and again with $x_2 = 1$.